Spam Message Detection using Machine Learning With User Interface:

1. Introduction

Spam messages are unwanted messages that can mislead users or cause harm. With the increasing use of mobile communication, detecting spam messages has become important. This project aims to build a machine learning model that can classify messages as spam or not spam.

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2. Problem Statement

The problem is to identify whether a given SMS message is spam or not using machine learning techniques.

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3. Objectives

- To build a model that detects spam messages
- To apply machine learning concepts learned in the course
- To improve accuracy in classification

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4. Methodology

The dataset used is the SMS Spam Collection dataset. The data contains labeled messages as spam or ham (not spam).

Steps followed:

- Data loading
- Data preprocessing
- Feature extraction using CountVectorizer
- Model training using Naive Bayes
- Model evaluation

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5. Tools and Technologies

- Python
- Pandas
- Scikit-learn
- Streamlit (for UI)
- Matplotlib (for graph visualization)

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6. Implementation

The dataset is read using pandas. The messages are converted into numerical form using CountVectorizer. A Naive Bayes model is trained on the dataset. The model predicts whether a message is spam or not.

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7. Results

The model achieved an accuracy of approximately 95–98%. The system correctly classifies spam and non-spam messages. The user interface allows real-time prediction, and graphical visualization improves understanding of the dataset.

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8. Challenges Faced

- Understanding text preprocessing
- Handling dataset format
- Running Python environment

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9. Conclusion

This project successfully demonstrates the use of machine learning for spam detection. It also includes a user-friendly interface, making it practical and interactive. The project helped in understanding text processing, classification algorithms, and real-world application of AI.

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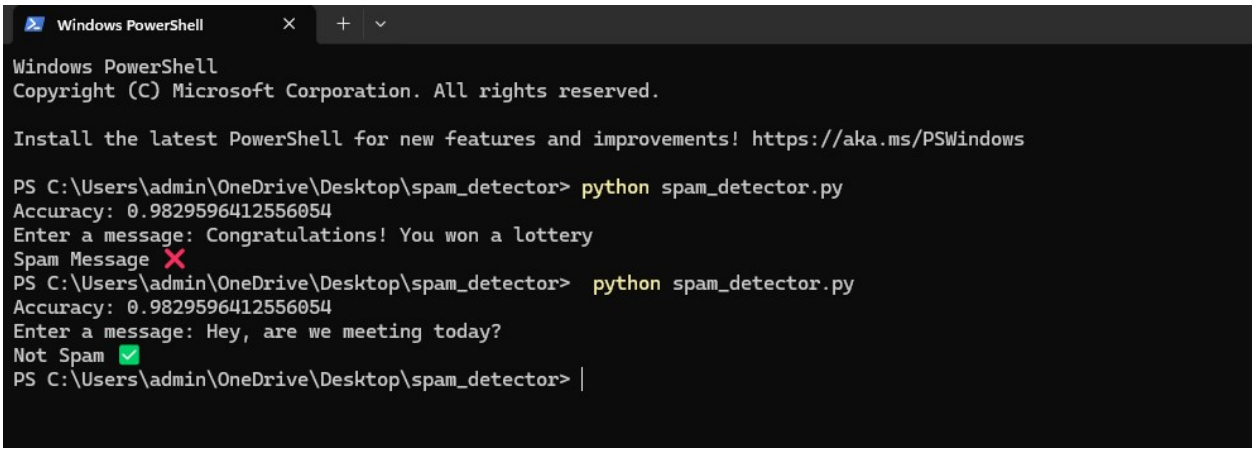
10. Future Scope

- Improve accuracy using advanced models
- Build a web or mobile application
- Use deep learning techniques

Output of Spam Detection Model:



```
Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\admin\OneDrive\Desktop\spam_detector> python spam_detector.py
Accuracy: 0.9829596412556054
Enter a message: Congratulations! You won a lottery
Spam Message ❌
PS C:\Users\admin\OneDrive\Desktop\spam_detector>  python spam_detector.py
Accuracy: 0.9829596412556054
Enter a message: Hey, are we meeting today?
Not Spam ✅
PS C:\Users\admin\OneDrive\Desktop\spam_detector>
```
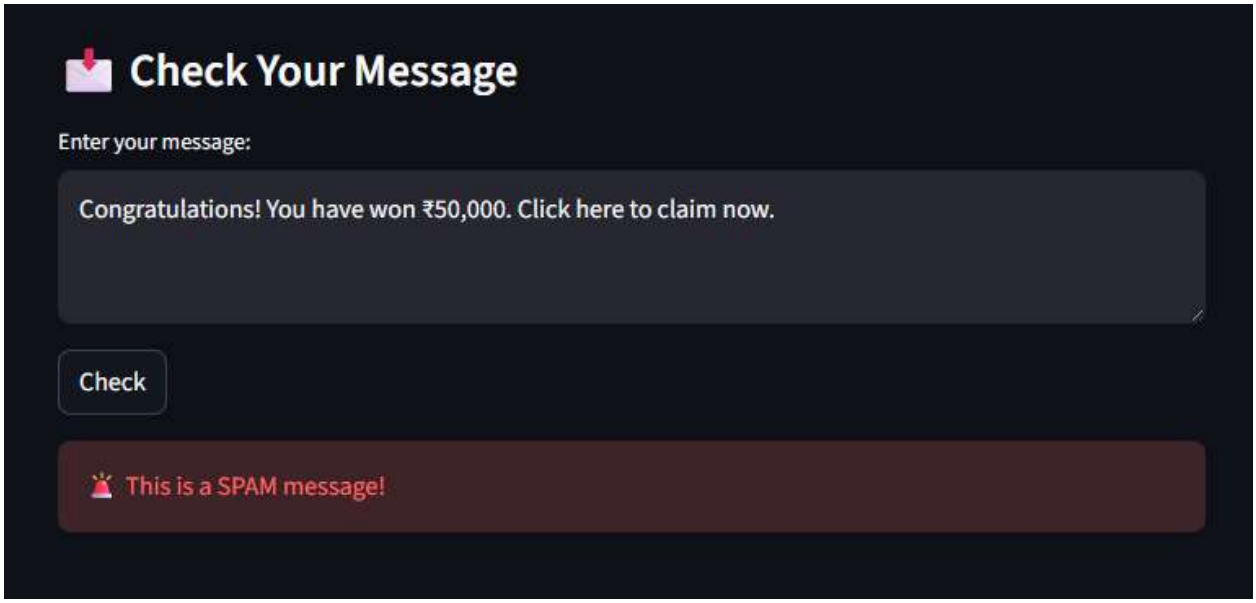

11. System Implementation

A Streamlit-based user interface was developed to make the model interactive. Users can input a message and instantly get prediction results. A graphical visualization is also included to show the distribution of spam and non-spam messages.

SCREENSHOT SECTION:

Figure 1: User Interface of Spam Detection System



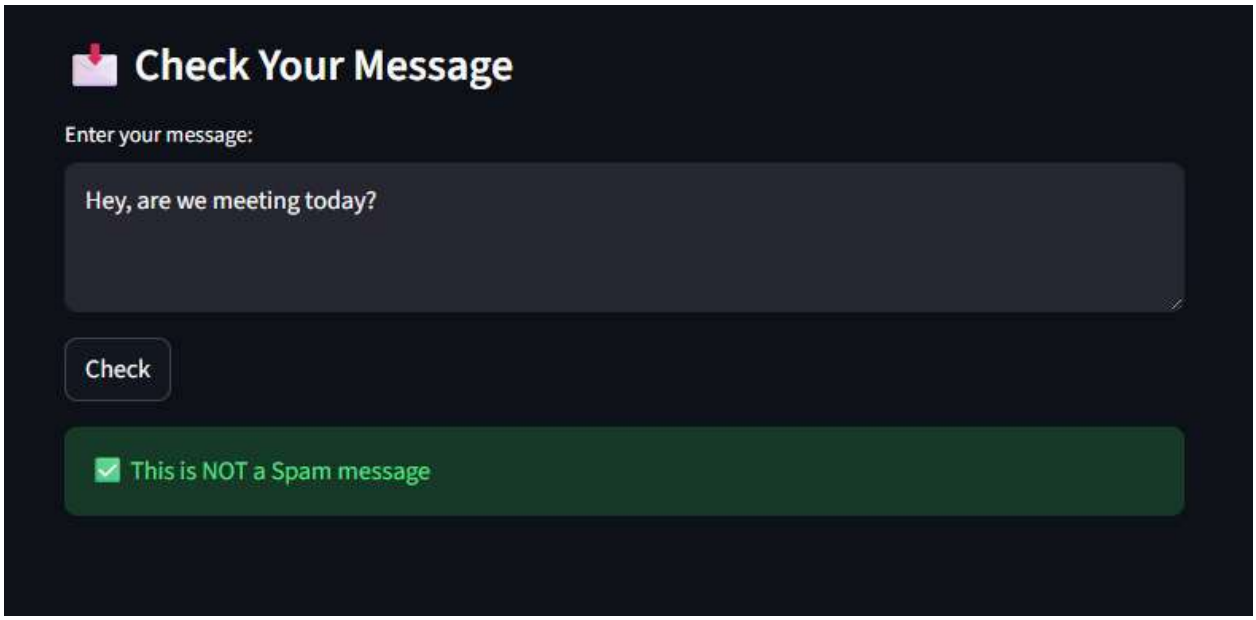
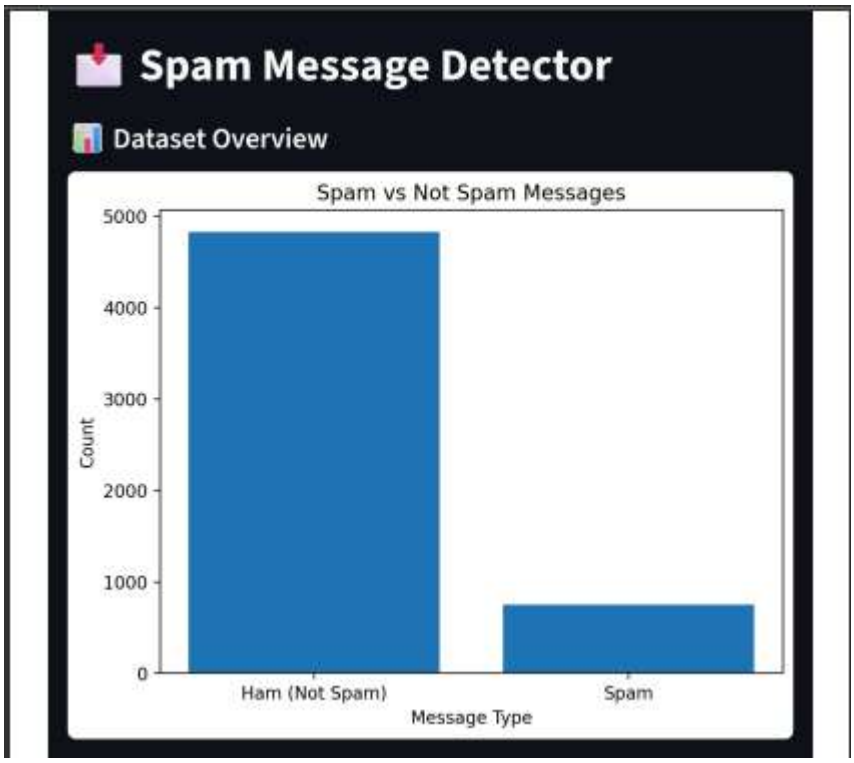



Figure 2: Graph showing Spam vs Non-Spam messages